

APPLIED CHEMISTRY

UNIT-5

NANO MATERIALS

BIT DURG



Department of Applied Chemistry

Bhilai Institute of Technology, Durg (C.G.)

B.Tech. First / Second Semester, BIT Durg, Applied Chemistry; Unit-5

SYLLABUS

UNIT – IV: Nanomaterials

CO5

Introduction, Top-down and Bottom-up approach, Classification on dimension, Characteristic, properties & application: Fullerenes, Carbon nanotube, Nanowire, Application of Nanomaterial: Catalysis, Medicine, Energy science, Bio nanomaterials. **[8Hrs]**

Course Outcome:

CO5: Explore the benefits and applications of nano materials.

UNIT-5

NANO MATERIALS

Introduction: Nanochemistry is the study of the fundamental principles of molecules and structures with at least one dimension between 1 and 100 nanometers. These structures are known as nanostructures. Nanotechnology is the application of these nanostructures into useful nanoscale devices. The difference between nano science chemistry and nano technology is that the former is related to the study of relationship between various phenomena, physical properties and material dimensions at nano-scale, whereas the later is application of the nano-structures and principles related to them to make nano-scale dimensions and material.

During the last few years, a little word with a big potential has diverted the attention of scientists all over the globe. That word is “nano”. Chemists, biologist, doctors, physicists, engineers, and computer scientists are all intimately involved in nano-development. Nanotechnology is the understanding and control of matter at the nanoscale, at dimensions between approximately 1 and 100 nanometers.

The prefix “nano” means one billionth. One nanometer is $1 / 1000,000,000$ of a meter i.e., one thousands millionth of a meter. To get a sense of the nano scale, a human hair measures 80,000 nanometers across, a bacterial cell measures a few hundred nanometers across, and the smallest features that are commonly etched on a commercial microchip are around 130 nanometers across. The smallest things observable with the unaided human eye are 10,000 nanometers across. Just ten hydrogen atoms or 5 silicon atoms in a line make one nanometer.

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It's really very small indeed. For comparison, a red blood cell is approximately 7000 nm wide and a water molecule is almost 0.3 nm across. Nanoscale is defined to be from 100 nm down to the size of atoms (approximately 0.2 nm) because it is at this scale that the properties of materials can be very different from those at a larger scale.

Nanoparticles:

- Nano particles are particles b/w 1 to 100 nm in size with a surrounding interfacial layers.
- The interfacial layer is an integral part of a nanoscale matter, fundamentally affecting all of its properties.
- The interfacial layer typically consists of ions, inorganic and organic molecules.
- Organic molecules coating inorganic nanoparticles are known as stabilizers, capping and surface ligands or passivating agents.
- In nanotechnology, a particle is defined as a small object that behaves as a whole unit with respect to its transport and properties.

Properties of nanoparticles:

- Nanoparticles act as bridge b/w bulk materials and atomic or molecular structures.
- The properties of the materials change as their size approaches the nanoscale dimensions.
- They also possess optical properties as they are small enough to confine their electrons and produce quantum effects.
- **Forex:** Gold nanoparticles appear deep-red to black in solution.
- Gold nanoparticles melt at much lower temperature ($\sim 300^{\circ}\text{C}$ for 2.5 nm size) than the gold slabs (1064°C).
- The high surface area to volume ratio of nanoparticles provides a tremendous driving force for diffusion.
- The materials composed of nanoparticles absorb solar radiation to a greater extent.
- Suspensions of nanoparticles are possible since the interaction of particle surface with the solvent is strong enough to overcome density differences.

Classification on Dimensions: This classification is based on the number of dimensions of a material, which are outside the nanoscale (<100 nm) range.

- In zero-dimensional (0D) nanomaterials all the dimensions are measured within the nanoscale (no dimensions are larger than 100 nm). Most commonly, 0D nanomaterials are nanoparticles.
- In one-dimensional nanomaterials (1D), one dimension is outside the nanoscale. This class includes nanotubes, nanorods, and nanowires.
- In two-dimensional nanomaterials (2D), two dimensions are outside the nanoscale. This class exhibits plate-like shapes and includes graphene, nanofilms, nanolayers, and nanocoatings.

Synthesis of nanoparticles: Since nanoparticles have unique property they are used in various antimicrobial applications, biomass or materials, composite fibres, cryogenic superconducting materials, cosmetic products and electronic components, hence find large application in pharmaceutical, medicine and dentistry.

The important chemical approaches for the synthesis of nano particles are:

- Top down and – bottom up

Top down:

- This method involves attrition or milling which involves a mechanical thermal cycle of preparing nanoparticle.
- This seeks to create smaller devices using larger ones to direct their assembly.
- The most common top down approach to fabrication involves lithographics patterning techniques using short wavelength optical sources.
- Top down approach refers to slicing or successive cutting of a bulk material to get nano sized particle.
- The starting material is solid state.
- It includes physical processing methods:
 - Mechanical methods:
 - Cutting, etching, grinding
 - Ball milling
 - Lithographic techniques:
 - Photolithography
 - e-beam lithography

- Its advantages are:
 - It provides large scale production.
 - Deposition over a large substrate is possible.
 - Chemical purification is not required.
- Its disadvantages are:
 - Yields:
 - Broad size distribution (10 – 1000 nm)
 - Varied particles shapes or geometry
 - Control over deposition parameters is difficult to achieve.
 - **Impurities:** stresses, defects and imperfections get introduced.

Bottom up:

- Bottom up approach refers to the buildup of a material from the bottom: atom by atom, molecule by molecule.
- Atom by atom deposition leads to formation of self assembly of atoms / molecules and clusters.
- These clusters come together to form assembled monolayers on the surface of substrate.
- The starting material is either gases state or liquid state of matter.
- It was chemical or physical forces operating at the nanoscale to assemble basic units into larger structures.
- **Ex.:** (a) Indium gallium overlayer (in GaAs) quantum dots can be formed by growing thin layers of InGaAs on Ga As.
- (b) Formation of carbon nanotubes.
- It includes physical and chemical processing method.
- **Physical techniques:**
 - Physical Vapour Deposition (PVD) involves condensation of vapour phase species.
 - Evaporation (Thermal e-beam)
 - Sputtering
 - Plasma Arcing
 - Laser ablation
- **Chemical techniques:**
 - CVD: Deposition of vapour phase of reactant species.

- **Self assembled monolayer:**
 - Electrolytic deposition
 - Solgel method
 - Pyrolysis
 - Micro emulsion route
- Its advantages are:
 - Ultra fine nanoparticles, nanoshells, nanotubes can be prepared.
 - Deposition parameters can be controlled
 - Cheaper technique
 - Narrow size distribution is possible (1 – 20 nm).
- Its disadvantages are:
- Large scale production is difficult.
- Chemical purification of nanoparticles is required.

Fullerenes (C₆₀): Fullerenes are family of carbon allotropes, i.e., molecules composed of entirely carbon, in the form of a hollow sphere, ellipsoid, tube or plane. Spherical fullerenes are also called buckyballs, and cylindrical ones are called carbon nanotubes or buckytubes. Graphene is an example of a planar fullerenes sheet. Its structure is one atom thick planner sheets of Sp²-bonded C atoms that are densely packed in a honeycomb crystal lattice. These include the carbon nanotubes (or silicon nanotubes), which are of interest both because of their mechanical strength and also because of their electrical properties.

Fullerenes are similar in structure to graphite, which is composed of sheets of hexagonal rings, but may also contain pentagon rings.

Fullerenes are defined as an elongated sphere of carbon atoms formed by interconnecting six-member rings and isolated five-member rings forming hexagonal and pentagonal faces. The first isolated and characterized fullerene, C₆₀, contains 20 hexagonal faces and 12 pentagonal faces just like a soccer ball and possess perfect symmetry.

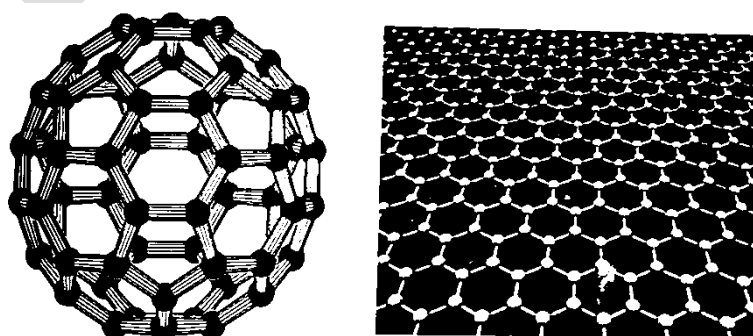
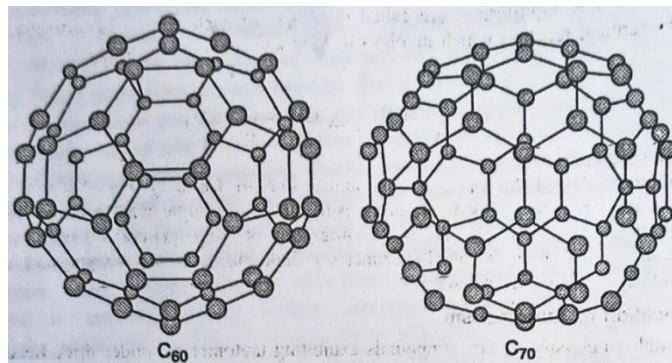


Fig. 1: (a) Buckminsterfullerene C₆₀ (b) Graphene lattice



Characteristics of Fullerenes:

1. They are insoluble in water, but soluble in organic solvents. C-60 is soluble in benzene and can be crystallized from it.
2. They exhibit excellent tensile strength and very high packing density.
3. C-60 is non-toxic and a powerful antioxidant. In living cells, it reacts with free radicals responsible for the damage.
4. They possess cage like structure.
5. They are impermeable to all elements under normal circumstances.
6. C-60 crystal is an insulator but when doped with an alkali atom, it becomes conducting.
7. C-60 crystal doped with potassium becomes superconductor at 15 K.

Applications of Fullerenes: Fullerene chemistry continues to be an exciting field generating many articles with promising no applications every year. Some examples are given below:

1. Magnetic nanoparticles show great potential for high-density magnetic storage media.
2. C-60 dispersed into ferromagnetic materials such as iron, cobalt can form thin films with promising magnetic properties.
3. A number of organometallic-fullerene compounds have recently been synthesized which are being used in industry.
4. Some fullerene derivatives even exhibit superconducting character. There has been a report of a fullerene containing, superconducting field-effect device with a T_c as high as 117 K.
5. Fullerenes are used as miniature ball bearings for lubrication.

Carbon Nanotubes (CNTs): Carbon nanotubes are hollow cylinders of carbon atoms. Their appearance is like rolled tubes of graphite. The ends of CNTs have domed structures of six-membered rings capped by a five-membered ring. Carbon nanotubes (CNTs) were first isolated and characterized by Iijima in 1991. Carbon nanotubes (CNTs) are tubes made of carbon with diameters typically measured in nanometers. Carbon nanotubes also often refer to multi-wall carbon nanotubes (MWCNTs) consisting of nested single-wall carbon nanotubes weakly bound together by van der Waals interactions in a tree ring-like structure. Carbon nanotubes can exhibit remarkable electrical conductivity, while others are semiconductors. These properties are expected to be valuable in many areas of technology, such as electronics, optics, composite materials (replacing or complementing carbon fibers), nanotechnology, and other applications of materials science.

There are two types of CNTs: Single-walled carbon nanotubes (SWCNTs) and multi-walled carbon nanotubes (MWCNTs). As their names imply, SWCNTs consist of a single, cylindrical graphene layer, whereas MWCNTs consist of multiple graphene layers fused into one another. A large number of articles have been published on CNTs and many new applications of CNTs are emerging every year.

Characteristics of Carbon nanotubes: The unique physical and chemical properties of CNTs, such as structural rigidity and flexibility, have increased their utility. Their important characteristics are given below:

1. CNTs are extremely strong, about 100 times stronger than steel.
2. CNTs can also act as either conductor or semiconductors depending on their chirality.
3. They can act as superconductors, and can also behave as field emitters.
4. They possess very low density about half of density of aluminium metal.
5. They possess high elasticity and high thermal conductivity almost twice as that of diamond.
6. They have high current density and high aspect ratio i.e., length to diameter ratio.

Applications:

1. Carbon nanotubes are now the top candidates to replace silicon based semiconductors as they have variable semiconducting properties with energy gaps ranging from a few milli electron volt (meV) to a few tenths of an electron volt (eV).
2. Nanotubes have led to a new generation of electronic devices.
3. Their use as ultra-sensitive electromechanical sensors has also been explored.

4. They are used in designing of field-effect transistors (FETs) and integrated electronic devices.
5. They are used in designing chemical and radiation sensors.
6. They are used in composites and manufacture of fuel cells.

Nanowire: Nanowires are just like normal electrical wires other than the fact that they are extremely small, Like conventional wires, nanowires can be made from a variety of conducting and semiconducting materials like copper, silver, gold, iron, silicon, zinc oxide and germanium. Nanowires can also be made from carbon nanotubes. Nanowires are less than 100 nanometers in diameter and can be as small as 3 nanometers. Typically nanowires are more than 1000 times longer than their diameter. This massive difference in the length to diameter ratio means that nanowires are often referred to as 1-dimensional materials. This leads to unique properties that are not seen in the bulk materials, such as Quantum Mechanical Effects.

Applications: Nanowires show promise for use in applications including.

- Exceptionally small electronic circuits
- Memory devices
- Transistors
- Advanced composite materials
- Quantum devices
- Biomolecular nanosensors
- MEMS
- Optoelectronics
- Field Emitters
- Photon Ballistic Waveguides

Applications of Nanomaterials:

1. **Catalysis:** Nanomaterial-based catalysts are homo and heterogeneous catalysts types. The extremely small size of the particles maximizes surface area exposed to the reactant, allowing more reactions to occur. However, thermal stability of these nanomaterials is limited by their critical sizes; the smaller the crystallite size, the lower the thermal stability.

Nanocatalysts as compared to their bulk counterparts, commonly offers much higher surface-to-volume ratio. Prominent changes in the electronic states and coordination environment of the surface atoms of a catalyst nanoparticle might be possible when its size decreases typically to a certain nanoregime. Therefore, change in size of nanoparticles affects coordination environment, electronic state, and adsorption energy of the reactant molecules. High surface area and consequently, enhanced surface active sites have led to extensive use of nanoparticles (NPs) as catalysts. Altering the nature and density of active sites can improve their catalytic activity. Change in nanoparticle size, shape, and composition affects the active site catalytic properties. Some properties of nano catalysts are:

- High activity
- High selectivity
- Excellent stability
- Easily separable
- Energy efficient

Applications of nano catalysts: Main applications of nanocatalysts are in

- water purification
- fuel cell
- energy storage
- in composite solid rocket propellants
- bio diesel production
- in medicine
- in dye

2. Medical Applications:

- Nanotechnology is already broadening the medical tools, knowledge, and therapies currently available to clinicians. Nano medicine, the application of nanotechnology in medicine, draws on the natural scale of biological phenomena to produce precise solutions for disease prevention, diagnosis, and treatment. Below are some examples of recent advances in this area: Commercial applications have adapted gold nano particles as probes for the detection of targeted sequences of nucleic acids, and gold nano particles are also being clinically investigated as potential treatments for cancer and other diseases.
- Better imaging and diagnostic tools enabled by nanotechnology are paving the way for earlier diagnosis, more individualized treatment options, and better therapeutic success rates.
- Nanotechnology researchers are working on a number of different therapeutics where a nanoparticle can encapsulate or otherwise help to deliver medication directly to cancer cells and minimize the risk of damage to healthy tissue. This has the potential to change the way doctors treat cancer and dramatically reduce the toxic effects of chemotherapy.
- Nanomedicine researchers are looking at ways that nanotechnology can improve vaccines, including vaccine delivery without the use of needles.

3. Energy Applications: Nanotechnology is finding application in traditional energy sources and is greatly enhancing alternative energy approaches to help meet the world's increasing energy demands. Many scientists are looking into ways to develop clean, affordable, and renewable energy sources, along with means to reduce energy consumption and lessen toxicity burdens on the environment:

- Nanotechnology is improving the efficiency of fuel production from raw petroleum materials through better catalysis. It is also enabling reduced fuel consumption in vehicles and power plants through higher-efficiency combustion and decreased friction.
- Nanotechnology is also being applied to oil and gas extraction through, for example, the use of nanotechnology-enabled gas lift valves in offshore operations or the use of nano particles to detect microscopic down-well oil pipeline fractures.

- Researchers are investigating carbon nanotube “scrubbers” and membranes to separate carbon dioxide from power plant exhaust.
- Researchers are developing wires containing carbon nano tubes that will have much lower resistance than the high-tension wires currently used in the electric grid, thus reducing transmission power loss.
- Nanotechnology can be incorporated into solar panels to convert sunlight to electricity more efficiently, promising inexpensive solar power in the future. Nano structured solar cells could be cheaper to manufacture and easier to install, since they can use print-like manufacturing processes and can be made in flexible rolls rather than discrete panels. Newer research suggests that future solar converters might even be “paintable.”
- Nanotechnology is already being used to develop many new kinds of batteries that are quicker-charging, more efficient, lighter weight, have a higher power density, and hold electrical charge longer.
- An epoxy containing carbon nanotubes is being used to make windmill blades that are longer, stronger, and lighter-weight than other blades to increase the amount of electricity that windmills can generate.
- In the area of energy harvesting, researchers are developing thin-film solar electric panels that can be fitted onto computer cases and flexible piezoelectric nano wires woven into clothing to generate usable energy on the go from light, friction, and / or body heat to power mobile electronic devices. Similarly, various nano science-based options are being pursued to convert waste heat in computers, automobiles, homes, power plants, etc., to usable electrical power.
- Energy efficiency and energy saving products are increasing in number and types of application. In addition to those noted above, nanotechnology is enabling more efficient lighting systems; lighter and stronger vehicle chassis materials for the transportation sector; lower energy consumption in advanced electronics; and light-responsive smart coatings for glass.

4. **Bio Nanomaterials:** The materials of biological origin are referred to as biological materials. These are used for nanotechnological applications. These materials comprised of biological molecules and have specific application which belongs to proteins, nucleic acids, lipids, hormones, vitamins, etc. Biological nanomaterials show properties such as: Self-assembling properties and specific molecular recognition.

DNA nanotechnology is one important example of bio nanotechnology. The utilization of the inherent properties of nucleic acids like DNA to create useful materials is a promising area of modern research. Another important area of research involves taking advantage of membrane properties to generate synthetic membranes. Proteins that self-assemble to generate functional materials could be used as a novel approach for the large-scale production of programmable nanomaterials. One example is the development of amyloids found in bacterial biofilms as engineered nanomaterials that can be programmed genetically to have different properties. Lipid nanotechnology is another major area of research in bio nanotechnology, where physico-chemical properties of lipids such as their antifouling and self-assembly is exploited to build nano devices with applications in medicine and engineering.

Applications of Biological Nanomaterials:

1. Biological nanomaterials are used for specific filtration and drug delivery.
2. They are used as sensors, motors and actuators.
3. They are also used in arrays (chips) for large scale protein or DNA screening.
4. These materials find application in the formation of memory devices.

Questions:

1. Explain fabrication technique of nanoparticle.
2. Give short notes on application of nanotechnology in the field of:
(i) Medicine (ii) Energy Science (iii) Catalysis
3. Give short notes on following:
(i) Fullerenes (ii) Carbon nanotubes (iii) Nanowires (iv) Bio-nanomaterials
4. Define nanotechnology.
5. Discuss in detail:
(i) Fullerenes
(ii) Nanotubes
(iii) Nanowires

Reference Book:

1. Introduction to Nanoscience S.M. Lindsay 1st Edition 2012 Oxford University Press.



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